

TFRank: Think-Free Reasoning Enables Practical Pointwise LLM Ranking

Yongqi Fan^{◇*}, Xiaoyang Chen^{♥♣*}, Dezhi Ye[♠], Jie Liu[♠], Haijin Liang[♠]
Jin Ma[♠], Ben He^{♥♣}, Yingfei Sun[♥], Tong Ruan^{◇†}

[◇]East China University of Science and Technology, Shanghai, China

[♠]Tencent, [♥]University of Chinese Academy of Sciences

[♣]Chinese Information Processing Laboratory, Institute of Software, Chinese Academy of Sciences
johnnyfans@mail.ecust.edu.cn, chenxiaoyang19@mails.ucas.ac.cn
dezhiye@tencent.com, {benhe, yfsun}@ucas.ac.cn, ruantong@ecust.edu.cn

Abstract

Reasoning-intensive ranking models built on Large Language Models (LLMs) have made notable progress, but existing approaches often rely on large-scale LLMs and explicit Chain-of-Thought (CoT) reasoning, resulting in high computational cost and latency that limit real-world use. To address this, we propose **TFRank**, an efficient pointwise reasoning ranker based on small-scale LLMs. To improve ranking performance, TFRank effectively integrates CoT data, fine-grained score supervision, and multi-task training. Furthermore, it achieves an efficient “Think-Free” reasoning capability by employing a “think-mode switch” and pointwise format constraints. Specifically, this allows the model to leverage explicit reasoning during training while delivering precise relevance scores for complex queries at inference without generating any reasoning chains. Experiments show that TFRank (e.g., 1.7B) achieves performance comparable to models with four times more parameters on the BRIGHT benchmark, and demonstrates strong competitiveness on the BEIR benchmark. Further analysis shows that TFRank achieves an effective balance between performance and efficiency, providing a practical solution for integrating advanced reasoning into real-world systems. Our code and data are released in the repository: <https://github.com/JOHNNY-fans/TFRank>.

1 Introduction

Driven by the advancements in Large Language Models (LLMs), Information Retrieval (IR) systems are increasingly confronted with the demand to handle complex, inferential user queries (DeepSeek-AI et al. 2025; Yang et al. 2025a; Su et al. 2025; Weller et al. 2025a). In modern applications such as Retrieval-Augmented Generation (RAG) systems and Multi-Agent frameworks (Yu et al. 2024; Chen et al. 2024b; Guo et al. 2024), the ability to perform reasoning over both queries and retrieved documents is essential for ranking models to deliver accurate and relevant information. This has led to research on *reasoning-intensive ranking*, which aims to develop ranking algorithms capable of understanding subtle relationships and accurately assessing the relevance of documents to complex information needs.

Early attempts to endow rankers with reasoning capabilities relied primarily on the power of LLMs. Methods leverage LLMs’ instruction-following abilities to flexibly judge query–document relevance (Sun et al. 2023; Ma et al. 2023; Yu et al. 2024). To further enhance performance on complex queries, many approaches explicitly incorporate the Chain-of-Thought (CoT) mechanism. By generating reasoning text before making a final relevance judgment, models such as Rank1 (Weller et al. 2025b), Rank-R1 (Zhuang et al. 2025), and REARANK (Zhang et al. 2025a) have demonstrated significant performance improvements in handling queries that require sophisticated reasoning.

However, the pursuit of performance often incurs prohibitive costs, hindering the transition from research prototypes to deployable systems. A number of state-of-the-art rankers rely on computationally and financially expensive models, whether proprietary (e.g., GPT-4) or massive open-source versions (e.g., Llama-3-70B-Instruct (Dubey et al. 2024)). Moreover, the reasoning process that boost performance are token-intensive, leading to significant inference latency. Finally, listwise or setwise approaches, adopted for cross-document comparison, exhibit limited scalability, rendering them unsuitable for low-latency, real-time ranking.

Therefore, as illustrated in Figure 1, the core challenge in designing a practical, reasoning-capable ranker is: **How can we leverage the advantages of reasoning without sacrificing efficiency and robustness in real-world scenarios?** A deployable ranker must meet two essential requirements. It should offer strong performance, including robust query–document relevance modeling, multi-document comparison, reasoning-based inference, and fine-grained scoring. It must also satisfy strict efficiency constraints, such as compact model size, short output length, stable formatting, and a scalable pointwise ranking architecture.

To address this challenge, we propose a novel training and inference pipeline for *pointwise* ranking that balances efficiency and performance with small-scale LLMs. In the training phase, we adopt a multi-task strategy that integrates pointwise, pairwise, and listwise supervision with fine-grained relevance labels and CoT signals, enabling the model to learn nuanced and precise scoring. To further strengthen the model’s capability, we incorporate an optional GRPO (Shao et al. 2024) training for advanced rea-

* Co-first authors, work was done when interning at Tencent.

† Corresponding author.

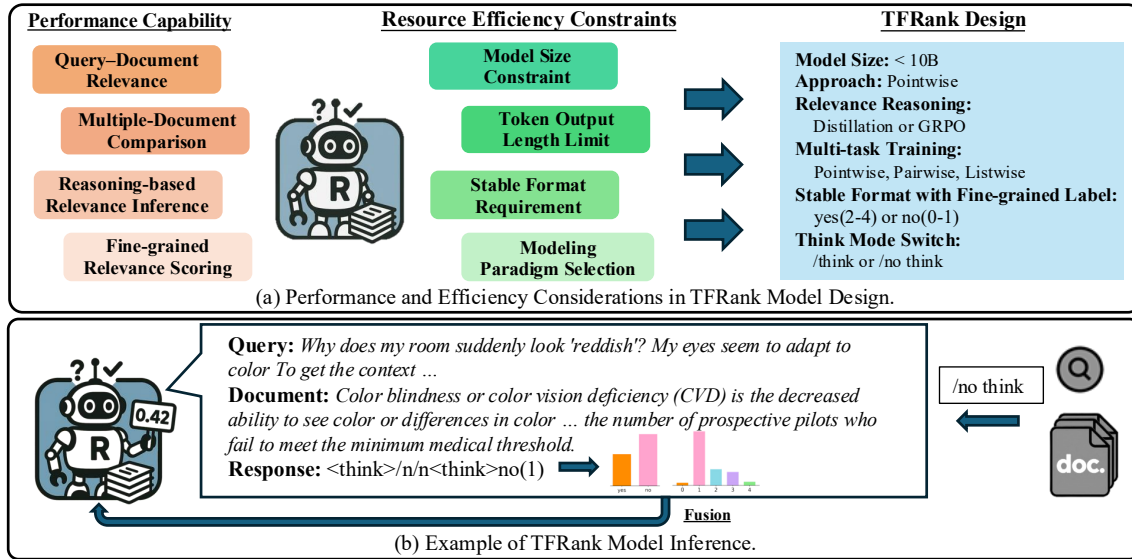


Figure 1: Overview of the TFRank framework. (a) summarizes key performance and efficiency considerations in the model design. (b) provides an example of inference with the TFRank model.

soning optimization. At inference, a “think-mode switch” and pointwise output constraints prompt the model to generate direct relevance scores for each query–document pair, fully bypassing explicit reasoning generation.

Experiments on the reasoning-intensive BRIGHT benchmark reveal a striking finding: our “Think-Free” inference mode, which omits explicit reasoning chains, not only dramatically improves efficiency but also surpasses the ranking performance of the full-reasoning mode. We term this phenomenon “**Think-Free** Reasoning” and name our model **TFRank** accordingly. This suggests the model internalizes its reasoning abilities via multi-task training, enabling it to produce high-quality judgments directly. The efficacy of TFRank is particularly pronounced on smaller models: for instance, our 1.7B TFRank model outperforms 7B-parameter baselines on BRIGHT, models over four times its size. Moreover, on the general BEIR benchmark, TFRank maintains comparable performance to existing methods while offering substantial efficiency gains. Thus, TFRank provides a validated and efficient pathway for building a new generation of ranking systems that combine state-of-the-art reasoning capabilities with production-level feasibility.

Our contributions can be summarized as follows:

- We propose **TFRank**, a novel and efficient framework that enables small-scale LLMs to perform reasoning-intensive ranking by internalizing reasoning patterns through multi-task training.
- We identify the “**Think-Free** Reasoning” phenomenon, where suppressing explicit reasoning chains at inference leads to both superior ranking accuracy and dramatic efficiency gains.
- Our experiments demonstrate that TFRank achieves promising results on the BRIGHT benchmark, with a 1.7B model outperforming 7B baselines, proving its

practical viability for production systems.

2 Related Works

Application of LLMs in Ranking LLM-based ranking methods can be categorized into three main paradigms based on how documents are processed. The **pointwise** paradigm evaluates each document independently for relevance to the query and outputs a score (Ma et al. 2024; Li et al. 2025; Weller et al. 2025b; Zhang et al. 2025b), offering high parallelism and low latency, thus making it suitable for real-time applications. The **pairwise** paradigm compares document pairs to provide finer-grained supervision (Qin et al. 2024; Luo et al. 2024), but its quadratic complexity limits scalability when ranking many candidates. **Listwise** and **setwise** methods (Sun et al. 2023; Ma et al. 2023; Zhuang et al. 2024) process multiple documents jointly, yet face challenges such as increased input/output length and lower efficiency due to sliding window strategies, making real-time deployment difficult. Our proposed TFRank adopts the efficient pointwise architecture, while integrating strong reasoning abilities through targeted training.

Reasoning-Intensive LLM Ranking With the rapid progress of RAG and Multi-Agent systems, and the emergence of stronger LLMs (e.g., DeepSeek-R1 (DeepSeek-AI et al. 2025), OpenAI o1), explicit reasoning is increasingly integrated into ranking models. There are two main directions: one is to directly leverage the zero-shot reasoning abilities of large LLMs (e.g., Llama-3.1-70B (Dubey et al. 2024), GPT-4o), as in JudgeRank (Niu et al. 2024) and InsertRank (Seetharaman, Dhole, and Bansal 2025). The other is to equip rankers with reasoning via model training. This includes CoT distillation, such as ReasoningRank (Ji et al. 2024), Rank1 (Weller et al. 2025b), and Rank-K (Yang et al. 2025b), and GRPO-based approaches like REAR-

Model		Approach	StackExchange								Coding		Theorem-based			Avg.
			Bio.	Earth.	Econ.	Psy.	Rob.	Stack.	Sus.	Leet.	Pony	AoPS	TheoQ.	TheoT.		
BM25	-	Pointwise	18.2	27.9	16.5	13.4	10.9	16.3	16.1	24.7	4.3	6.5	7.3	2.1	13.7	
RankGPT-4	Zero-shot	Listwise	33.8	34.2	16.7	27.0	22.3	27.7	11.1	3.4	15.6	1.2	0.2	8.6	17.0	
Rank1-7B	SFT	Pointwise	31.4	36.7	18.3	25.4	13.8	17.6	24.8	16.7	9.5	6.1	9.5	<u>11.6</u>	18.5	
Rank1-14B	SFT	Pointwise	29.6	34.8	17.2	24.3	18.6	16.2	24.5	17.5	14.4	5.5	9.2	<u>10.7</u>	18.5	
REARANK-7B	GRPO	Listwise	23.4	27.4	18.5	24.2	17.4	16.3	<u>25.1</u>	<u>27.0</u>	8.0	7.4	7.9	9.5	17.7	
Rank-R1-3B	GRPO	Setwise	18.4	17.1	13.7	16.9	9.0	10.0	16.5	11.1	4.7	3.5	3.2	5.9	10.8	
Rank-R1-7B	GRPO	Setwise	26.0	28.5	17.2	24.2	19.1	10.4	24.2	19.8	4.3	4.3	8.3	10.9	16.4	
Rank-R1-14B	GRPO	Setwise	31.2	38.5	21.2	26.4	<u>22.6</u>	18.9	27.5	20.2	9.2	<u>9.7</u>	9.2	11.9	20.5	
Llama3.2 Series																
TFRank-1B	SFT	Pointwise	27.7	29.7	9.5	18.1	11.6	10.5	10.7	9.0	10.4	5.9	5.9	1.5	12.5	
TFRank-3B	SFT	Pointwise	<u>34.9</u>	<u>46.1</u>	19.7	27.2	16.6	18.6	22.4	20.6	2.8	8.1	9.3	8.2	19.5	
TFRank-8B	SFT	Pointwise	35.2	44.6	<u>21.9</u>	30.7	21.0	21.0	23.3	26.6	18.3	12.6	9.6	9.1	22.8	
Qwen2.5 Series																
TFRank-0.5B	SFT	Pointwise	18.6	23.0	7.5	14.8	16.4	8.3	13.1	8.6	<u>16.1</u>	4.0	3.0	1.0	11.2	
TFRank-1.5B	SFT	Pointwise	25.9	36.7	17.3	21.3	16.2	14.4	15.7	17.0	13.4	7.2	9.6	6.3	16.8	
TFRank-3B	SFT	Pointwise	32.3	41.2	17.6	23.1	17.1	15.9	21.2	24.9	12.0	<u>9.7</u>	9.3	8.3	19.4	
TFRank-7B	SFT	Pointwise	34.3	45.2	20.5	26.0	22.4	19.1	24.6	24.9	8.9	6.8	9.0	9.7	21.0	
Qwen3 Series																
TFRank-0.6B	SFT	Pointwise	25.4	34.6	16.6	19.8	19.1	14.1	18.5	26.3	15.9	8.4	10.7	9.6	18.3	
TFRank-1.7B	SFT	Pointwise	29.9	38.7	18.2	<u>29.6</u>	17.3	16.4	18.9	22.4	11.0	5.2	6.9	9.3	18.7	
TFRank-4B	SFT	Pointwise	31.7	45.9	20.3	27.2	21.7	19.6	20.5	23.6	8.7	8.0	10.1	9.2	20.5	
TFRank-8B	SFT	Pointwise	33.0	47.9	24.6	27.4	22.8	<u>21.2</u>	24.1	28.5	9.8	9.0	<u>10.6</u>	9.1	<u>22.3</u>	

Table 1: Main evaluation results (NDCG@10) on the BRIGHT benchmark using BM25 as retriever. TFRank models are trained on their respective backbones. For each column, the highest value is **bolded** and the second highest is underlined.

ANK (Zhang et al. 2025a) and Rank-R1 (Zhuang et al. 2025), which optimize reasoning-based ranking on lists or sets (Shao et al. 2024). However, most existing methods rely on large-scale models and online CoT generation, resulting in high inference costs that hinder scalable deployment. In contrast, TFRank is built on small-scale LLMs and uses CoT only as a supervision signal during training. During inference, it fully bypasses CoT via a “Think-Free” mechanism, achieving high efficiency without sacrificing performance.

3 Methods

3.1 Overall Architecture

TFRank employs a two-stage architecture. In the *training stage*, we instill reasoning abilities into a small-scale LLM (0.5~8B) through multi-task supervised fine-tuning (SFT). This process leverages a diverse dataset labeled with binary relevance, fine-grained scores, and explicit CoT reasoning. An optional GRPO-based optimization step can also be applied to further enhance the model’s performance. During the *inference stage*, inspired by the Qwen3 series (Yang et al. 2025a), we employ a special prompt token as the “think-mode switch” to force the model to bypass explicit CoT generation and emit only a structured pointwise relevance score, thus minimizing latency and output length, making it suitable for real-time ranking scenarios.

3.2 Training Data Construction

High-quality training data is the foundation of TFRank’s performance. Our data construction involves two steps:

Construction of Foundational Relevance Datasets To equip the model with fundamental ranking abilities, we first constructed two datasets containing binary and fine-grained relevance labels, respectively. For **binary relevance data**, following the data processing methodology of Rank1, we acquire approximately 386k positive and negative query-document pairs from the MS MARCO dataset (Nguyen et al. 2016), using only their binary labels; this dataset is denoted as `ms_sub_binary`. For **fine-grained score data**, to provide the model with more nuanced judgment capabilities, we follow the methodology of BGE-M3 (Chen et al. 2024a). Concretely, we randomly sample approximately 10% of queries from each length derived from MS MARCO, including their top-1 positive and negative documents. For these pairs, we employ Deepseek-R1 to generate fine-grained relevance scores on a five-point scale. To ensure annotation quality, we apply strict filtering: only positive samples with final scores in $[2, 3, 4]$ and negative samples with scores in $[0, 1]$ are retained. This process yields a fine-grained dataset, denoted as `ms_sub_finegrained`, comprising approximately 7k queries and 156k query-document pairs. A more detailed breakdown of the training data categories is provided in the supplementary materials.

Model		Approach	Avg.
BM25	-	Pointwise	36.9
Rank1-7B	SFT	Pointwise	39.2
Rank1-14B	SFT	Pointwise	38.7
REARANK-7B	GRPO	Listwise	42.8
Rank-R1-3B	GRPO	Setwise	37.8
Rank-R1-7B	GRPO	Setwise	43.5
Rank-R1-14B	GRPO	Setwise	43.8

Llama3.2 Series			
TFRank-1B	SFT	Pointwise	29.9
TFRank-3B	SFT	Pointwise	40.5
TFRank-8B	SFT	Pointwise	43.2

Qwen2.5 Series			
TFRank-0.5B	SFT	Pointwise	36.6
TFRank-1.5B	SFT	Pointwise	39.5
TFRank-3B	SFT	Pointwise	39.7
TFRank-7B	SFT	Pointwise	41.5

Qwen3 Series			
TFRank-0.6B	SFT	Pointwise	39.0
TFRank-1.7B	SFT	Pointwise	39.4
TFRank-4B	SFT	Pointwise	42.5
TFRank-8B	SFT	Pointwise	42.1

Table 2: Short evaluation results (average NDCG@10) on the BEIR benchmark, using BM25 as retriever. TFRank models are trained on each corresponding backbone. The best average value for each block is **bolded**, and the second best is underlined. Detailed results for all BEIR datasets are provided in the supplementary material.

Multi-task Data and CoT Distillation After constructing the `ms_sub_finegrained` dataset, we expand it into three task paradigms: **pointwise**, **pairwise**, and **listwise**. For pointwise data, we prompt Deepseek-R1 to generate a CoT explanation for each `[query, doc]` pair to justify its known fine-grained score, with the final label appended to the end. For the pairwise setting, we sample `[query, doc1, doc2]` triplets sharing the same query, and use Deepseek-R1 to generate a CoT explaining their relative relevance and to produce a preference label. For listwise tasks, we follow the REARANK (Zhang et al. 2025a) format: for each query, we construct a set of documents $D = \{d_1, d_2, \dots, d_n\}$, ensuring diversity by sampling no more than two documents per relevance level, and instruct Deepseek-R1 to generate a CoT that explains the overall ranking logic for the predefined order. This procedure results in a multi-task dataset (`ms_sub_finegrained_cot`) enriched with CoT reasoning signals across all paradigms.

3.3 SFT Training: Internalizing Reasoning

During the SFT stage, our primary goal is to internalize CoT reasoning capabilities into a small-scale model and teach it to switch between `/think` and `/no think` modes based on instructions.

Think-Mode Switch We control the model’s generation mode by appending a special token after the input content:

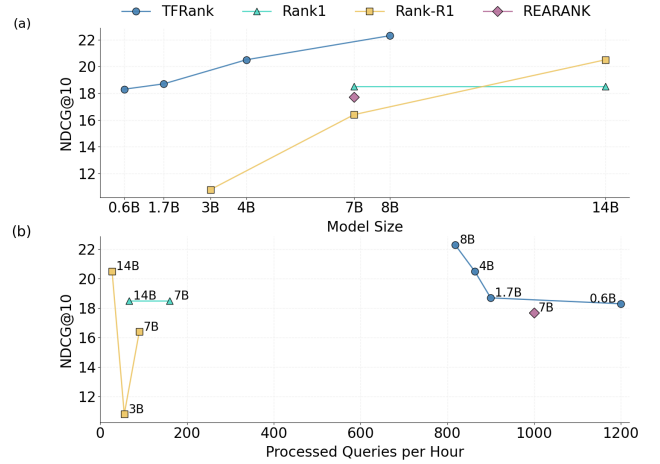


Figure 2: Size and efficiency trade-offs for ranking performance on the BRIGHT benchmark. (a) NDCG@10 versus model size for different ranker families; (b) NDCG@10 versus processed queries per hour (efficiency). All TFRank models are trained on the Qwen3 series.

- `/think`: Instructs the model to enter `/think` mode, requiring it to first generate a reasoning process in the format `<think>\n{reasoning text}\n</think>` before providing the final answer.
- `/no think`: Instructs the model to enter `/no think` mode, bypassing the reasoning process entirely and directly outputting the final answer after an empty `<think>\n\n</think>` tag.

Multi-task Supervised Fine-Tuning Based on different task paradigms (listwise, pairwise, and pointwise), constructed training datasets, and the think-mode switch, we construct a unified training data format for SFT:

- `<Instruction><Query><Doc><think mode>`

For clarity, we present below an example of a fine-grained pointwise sample in the `/no think` mode:

Prompt: `<Instruct>`: Please judge the relevance strength between the query and the document, and directly output the relevance judgment (yes or no), followed by the relevance score in parentheses, e.g., `yes(score)` or `no(score)`.
`<Query>`: what is a stereo preamplifier?
`<Doc>`: Amplifiers are essential components in any sound system, boosting the audio signal to drive loudspeakers and produce audible sound.
`/no think`

Response: `<think>\n\n</think>no (1)`

More detailed prompt and response formats for each paradigm are provided in the supplementary materials.

Through this hybrid training strategy, the model not only learns how to perform explicit reasoning but also learns to decouple its reasoning ability from the final judgment and switch its behavior based on instructions. The training loss for the SFT stage is the standard auto-regressive language

Model	SFT Setting	Inference Mode	StackExchange								Coding		Theorem-based			Avg.
			Bio.	Earth.	Econ.	Psy.	Rob.	Stack.	Sus.	Leet.	Pony	AoPS	TheoQ.	TheoT.		
Qwen3-0.6B	Zero-shot	w/o think	11.4	14.8	7.6	9.2	10.7	6.9	14.7	6.0	3.1	3.9	3.0	1.3	7.7	
TFRank-0.6B	full	w/o think	25.4	34.6	16.6	19.8	19.1	14.1	18.5	26.3	15.9	8.4	10.7	9.6	18.3	
	full	w/ think	12.7	21.1	10.0	15.1	4.4	8.6	10.2	14.1	4.9	3.8	5.9	3.8	9.5	
	w/o RR	w/o think	21.7	29.5	17.2	21.1	16.5	17.3	17.8	25.8	6.2	4.7	9.7	7.8	16.3	
	w/o FG	w/o think	27.9	34.0	14.8	22.2	20.5	12.3	15.5	23.9	2.9	7.5	10.5	10.0	16.8	
	w/o MT	w/o think	26.4	34.3	13.7	18.6	15.0	12.5	14.4	21.1	5.0	6.6	10.4	8.8	15.6	

Table 3: Ablation results (NDCG@10) of TFRank-0.6B (Qwen3) on BRIGHT using BM25 as retriever. “full” is the complete SFT pipeline; “w/ think” and “w/o think” denote inference with/without explicit reasoning. “w/o RR” is without Relevance Reasoning; “w/o FG” is without Fine-grained Label; “w/o MT” is without Multi-task.

model cross-entropy loss:

$$\mathcal{L}_{\text{SFT}} = - \sum_{i=1}^{|T|} \log P(t_i | t_{<i}, C) \quad (1)$$

where C is the input prompt and $T = \{t_1, \dots, t_{|T|}\}$ is the target sequence.

3.4 Inference: “Think-Free” Pointwise Ranking

During the inference stage, TFRank’s design is entirely guided by efficiency and practicality.

Activating /no think Mode For all inference requests, we uniformly append the `/no think` command to the prompt. This forces the model to suppress explicit CoT generation, causing it to output only a short answer containing the relevance judgment and score, such as `yes (4)` or `no (1)`. This approach reduces the generation length from hundreds of tokens to just a few, achieving an order-of-magnitude reduction in latency.

Pointwise Score Extraction and Fusion The model’s raw output contains two dimensions of signals: a binary judgment (yes/no) and a five-class fine-grained score (0-4). To obtain an expressive final ranking score, we designed a fusion formula. First, we calculate the probability of the binary judgment being “yes”, P_{bi} , from the model’s output logits:

$$P_{bi} = \text{softmax}(\text{logits}_{\text{yes}}, \text{logits}_{\text{no}}) \quad (2)$$

where $\text{logits}_{\text{yes}}$ and $\text{logits}_{\text{no}}$ are the model’s predicted logits for the “yes” and “no” tokens, respectively.

Next, we compute the expected value of the fine-grained score S_{fg} :

$$S_{fg} = \frac{\sum_{i=0}^4 i \cdot \text{softmax}(\text{logits}_{0\dots4})}{\max(i) - \min(i)} \quad (3)$$

where $\text{logits}_{0\dots4}$ are the model’s predicted logits for the five score tokens “0” through “4”.

Finally, we average these two scores to obtain the final TFRank sorting score:

$$\text{Score}_{\text{TFRank}} = 0.5 \cdot P_{bi} + 0.5 \cdot S_{fg} \quad (4)$$

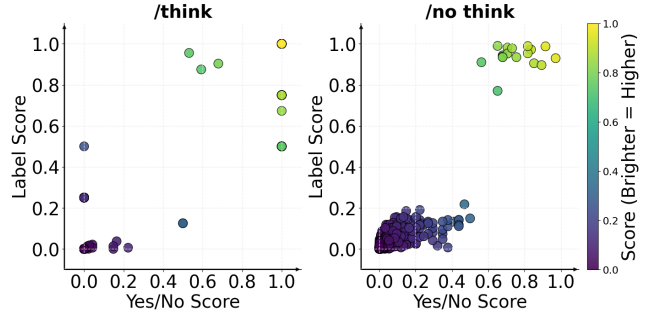


Figure 3: Score distributions for a random 1% sample of BRIGHT, evaluated by TFRank-0.6B (Qwen3) under `/think` and `/no think` inference modes.

3.5 Optional: Optimization with GRPO

To further explore the upper bound of relevance modeling in TFRank, we employ GRPO as an advanced optimization strategy. We compare two paradigms: **SFT-then-GRPO**, where the model first undergoes multi-task SFT before further GRPO optimization (labeled as **SFT+GRPO** in Table 4); and **Direct GRPO**, where end-to-end GRPO is directly applied to the instruction-tuned base model (GRPO). Considering the substantial training cost of GRPO, we randomly sample approximately 20% of queries based on the m3-data distribution, resulting in a GRPO training set with 8,641 samples and 1,348 unique queries.

We design a multi-dimensional reward function R_{total} to optimize both ranking quality and output consistency:

$$R_{\text{total}} = R_{\text{content}} + \lambda R_{\text{format}} \quad (5)$$

where λ is a balancing hyperparameter (default: 0.5).

Content Reward (R_{content}): The content reward is tailored for each task paradigm: for *pointwise* tasks, it averages binary classification accuracy and the mean squared error (MSE) of relevance scores; for *pairwise*, it averages preference accuracy and the MSE scores; for *listwise*, it is the mean of ranking relation accuracy, MSE scores, and relative nDCG improvement (as defined in REARANK).

Format Reward (R_{format}): The format reward enforces strict output conventions, including proper use of the `<think>` tag, minimum reasoning length in `/think`

Model		StackExchange							Coding		Theorem-based			Avg.
		Bio.	Earth.	Econ.	Psy.	Rob.	Stack.	Sus.	Leet.	Pony	AoPS	TheoQ.	TheoT.	
TFRank-0.6B	SFT	25.4	34.6	16.6	19.8	19.1	14.1	18.5	26.3	15.9	8.4	10.7	9.6	18.3
	GRPO	22.1	31.6	12.7	18.2	22.4	14.9	15.6	23.6	8.5	6.3	8.1	6.6	15.9
	SFT+GRPO	26.3	35.6	15.8	18.7	16.6	16.9	14.7	25.8	7.4	7.2	10.1	8.0	16.9
	GRPO [‡]	32.1	43.8	16.4	22.7	23.2	16.9	19.1	25.0	12.9	5.1	8.2	5.8	19.3
	SFT+GRPO [‡]	26.8	35.6	20.3	24.9	18.2	18.1	18.8	28.5	6.0	6.0	9.9	7.9	18.4
TFRank-1.7B	SFT	29.9	38.7	18.2	29.6	17.3	16.4	18.9	22.4	11.0	5.2	6.9	9.3	18.7
	SFT+GRPO	27.4	36.6	16.6	27.7	18.3	16.9	18.4	26.9	10.6	6.5	11.0	8.3	18.8
TFRank-4B	SFT	31.7	45.9	20.3	27.2	21.7	19.6	20.5	23.6	8.7	8.0	10.1	9.2	20.5
	SFT+GRPO	33.2	45.9	17.6	29.5	21.0	20.9	18.3	25.0	9.1	9.5	9.8	7.3	20.6
TFRank-8B	SFT	33.0	47.9	24.6	27.4	22.8	21.2	24.1	28.5	9.8	9.0	10.6	9.1	22.3
	SFT+GRPO	33.7	46.2	23.7	26.0	24.1	20.1	23.6	28.8	12.5	10.8	11.4	9.7	22.6

Table 4: Results (NDCG@10) of TFRank (Qwen3) on BRIGHT under different training paradigms, using BM25 as retriever. Models marked with [‡] apply GRPO using the full SFT training set.

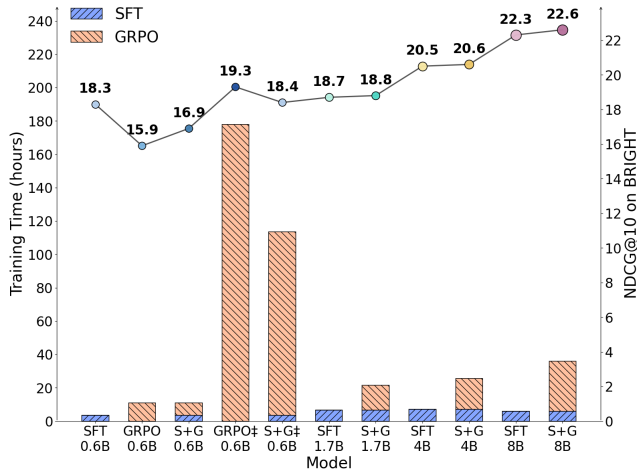


Figure 4: Training time (hours) versus ranking performance (NDCG@10 on BRIGHT) for different training strategies. All TFRank models use the Qwen3 series backbone.

mode, and adherence to the answer format required by each paradigm (e.g., *yes/no (score)* for pointwise).

Full reward definitions and implementation details are provided in the Supplementary Materials and released code.

4 Experiments

Datasets and Metrics We evaluate TFRank on two representative benchmarks: BRIGHT (Su et al. 2025) and BEIR (Thakur et al. 2021). BRIGHT contains approximately 1,384 real-world queries from diverse domains such as coding, mathematics, and economics, specifically designed to assess multi-step reasoning capabilities beyond keyword matching. BEIR comprises 18 public datasets covering a wide variety of retrieval tasks, including question answering, argument retrieval, fact-checking, biomedical search, and duplicate question detection, emphasizing zero-shot generalization across unseen domains. Following prior work,

we primarily report NDCG@10, which effectively measures ranking quality at top positions.

Baselines We compare TFRank against several representative ranking methods. BM25 serves as the classical lexical baseline. For LLM-based ranking, we include RankGPT (Sun et al. 2023), a GPT-4-based zero-shot list-wise reranker. Recent state-of-the-art methods considered are Rank1 (Weller et al. 2025b), Rank-R1 (Zhuang et al. 2025), and REARANK (Zhang et al. 2025a). All baselines are evaluated using official implementations or publicly released checkpoints.

Implementation Details TFRank models are trained on three LLM families: Qwen2.5 (0.5B~7B) (Yang et al. 2024), Qwen3 (0.6B~8B) (Yang et al. 2025a), and Llama 3.2 (1B~8B) (Dubey et al. 2024). Among these, Llama-3.2-8B is sourced from the open-source community, while other models are obtained from official repositories. Training uses a learning rate of 1×10^{-5} , random seed 42, and spans 5 epochs of supervised fine-tuning (SFT). In the GRPO optimization stage, we generate 8 completions per instance. For model checkpoint selection, 1% of the training data is set aside as a validation set. All experiments are conducted on a single 8-GPU H20 server, with inference batching supporting */think* and */no think* modes for efficiency. Typical model training completes within approximately 48 hours.

5 Results and Analysis

5.1 Main Results

Overall Performance As illustrated in Table 1, on the reasoning-intensive BRIGHT benchmark, TFRank **consistently and significantly outperforms** all existing baselines across multiple model families. Notably, TFRank unleashes the potential of small-scale LLMs. For example, Qwen3-based TFRank-1.7B achieves an average NDCG@10 of 18.7, surpassing the **4x**-larger Rank1-7B (18.5), REARANK-7B (17.7), and Rank-R1-7B (16.4). Likewise, TFRank-3B, based on Llama-3.2 and Qwen-2.5, also outperforms all the aforementioned 7B baselines. On

Retriever	Ranker	Score
ReasonIR-8B	Rank-R1-32B (Qwen2.5)	38.8
ReasonIR-8B	QwenRerank-32B (Qwen2.5)	36.9
ReasonIR-8B	TFRank-8B (Qwen3) SFT+GRPO	31.2
BM25 (pyserini)	Llama-Reasoning-70B (Llama-3.1)	25.7
ReasonIR-8B	REARANK-7B (Qwen2.5)	24.2
ReasonIR-8B	RankR1-7B (Qwen2.5)	23.2
BM25 (pyserini)	TFRank-8B (Qwen3) SFT+GRPO	22.6
BM25 (pyserini)	GritLM-7B (Mistral)	19.4
BM25 (pyserini)	RankGPT4 (gpt-4-0125-preview)	17.0

Table 5: BRIGHT leaderboard (NDCG@10), comparing advanced retriever-ranker pipelines.

the general-purpose BEIR benchmark, in Table 2, TFRank delivers performance on par with the strongest baselines. For instance, TFRank-8B, based on Llama-3.2, achieves an average NDCG@10 of 43.2, outperforming Rank1-14B and REARANK-7B, while remaining comparable to the 43.5 and 43.8 achieved by Rank-R1-7B and Rank-R1-14B.

Model Efficiency While achieving high performance, TFRank demonstrates **significant efficiency advantages**. As shown in Figure 2(a), on the BRIGHT benchmark, performance curve of TFRank is consistently above other models, indicating its ability to achieve superior performance at equivalent or even smaller model scales, which is crucial for resource-constrained deployment scenarios. Figure 2(b) further reveals its inference efficiency. By adopting a pointwise scoring and “Think-Free” reasoning mode, TFRank processes far more queries per hour than methods that require explicit CoT generation, such as Rank1 and Rank-R1. Compared to the listwise model REARANK, TFRank’s pointwise scoring enables intra-query parallelism, allowing single-query throughput to scale effectively with hardware resources. This is particularly advantageous for latency-sensitive applications. Remarkably, the 0.6B TFRank surpasses the 7B REARANK in performance on BRIGHT, while offering a significant throughput advantage.

5.2 Ablation Study

We conducted a series of ablation studies on the TFRank-0.6B (Qwen3) model, with the results presented in Table 3.

Effectiveness of Components Our full TFRank model achieves a performance of 18.3, more than doubling the score attained by its base model and demonstrating the overall effectiveness of our training method. Ablation experiments confirm that each of our proposed components is indispensable: removing either Relevance Reasoning Supervision, Fine-grained Relevance Labels, or Multi-task training results in a performance drop of over 8%.

“Think-Free” Reasoning A striking finding is the effectiveness of “Think-Free” reasoning. Forcing explicit CoT at inference (“w/ think”) severely damages performance, reducing NDCG@10 to 9.5. As illustrated in Figure 3, we observe that the score distribution in the /think mode is more concentrated, indicating a weaker capacity to discern

documents with partial relevance. This aligns with observations by Jedidi et al. (2025): explicit reasoning can induce overconfidence and impair the model’s ability to model partial relevance finely. By internalizing the reasoning process during training, TFRank can output more precise and discriminative scores at inference time without explicit “thinking”, thereby achieving a substantial increase in efficiency.

5.3 Further Optimization with GRPO

To explore the performance ceiling of TFRank, we introduced the GRPO training strategy. As shown in Table 4, for models of 1.7B and larger, applying GRPO training with a small amount of data after full SFT (SFT+GRPO) yields limited but consistent performance gains. Interestingly, on the 0.6B model, directly training with the full dataset using GRPO results in better performance (19.3 on NDCG@10) than first performing full SFT followed by full GRPO (SFT+GRPO[‡], 18.4). This may be because GRPO can explore a broader policy space than SFT, thus finding a better optimization path for smaller models. However, this performance improvement comes at a significant training cost. As depicted in Figure 4, the training duration for GRPO far exceeds that of SFT. For example, the cost of full GRPO training on the 0.6B model (approx. 200 hours) is much higher than that of training a superior-performing 8B SFT model (approx. 8 hours). Therefore, while increasing model scale via SFT or increasing the amount of GRPO data can both effectively boost performance, they represent a trade-off between computational resources and time.

5.4 BRIGHT Leaderboard

To validate TFRank’s competitiveness in state-of-the-art retrieval systems, we employ ReasonIR-8B (Shao et al. 2025) as the retriever and compare TFRank-8B, REARANK-7B, and Rank-R1-7B against top BRIGHT leaderboard entries. As shown in Table 5, the TFRank-8B model achieved an NDCG@10 score of 31.2, significantly surpassing REARANK-7B (24.2) and Rank-R1-7B (23.2). Although its performance is slightly lower than that of larger-scale models (e.g., Rank-R1-32B), TFRank’s results demonstrate that it is a highly competitive alternative in scenarios where deploying ultra-large models is not feasible.

6 Conclusion

We introduce TFRank, an innovative framework designed to address the efficiency bottlenecks of reasoning-intensive rankers in real-world deployments. Through our novel “Think-Free” reasoning mechanism, TFRank leverages multi-task learning and CoT data during training to internalize complex reasoning capabilities within small-scale LLMs. During inference, it completely bypasses the generation of explicit reasoning chains, directly and efficiently producing pointwise relevance scores. Experiments demonstrate that this approach enables small-scale models (e.g., 1.7B) to outperform baselines with over four times the parameters, while substantially reducing inference latency. We also uncover a key phenomenon: for models that have sufficiently internalized reasoning logic, suppressing explicit

CoT generation at inference time yields more accurate ranking results. In summary, TFRank presents a practical path for building and deploying advanced reasoning-based rankers, offering a feasible solution to popularize complex AI capabilities in resource-constrained applications.

References

- Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; and Liu, Z. 2024a. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *CoRR*, abs/2402.03216.
- Chen, X.; He, B.; Lin, H.; Han, X.; Wang, T.; Cao, B.; Sun, L.; and Sun, Y. 2024b. Spiral of Silence: How is Large Language Model Killing Information Retrieval? - A Case Study on Open Domain Question Answering. In Ku, L.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, Bangkok, Thailand, August 11-16, 2024, 14930–14951. Association for Computational Linguistics.
- DeepSeek-AI; Guo, D.; Yang, D.; Zhang, H.; Song, J.; Zhang, R.; Xu, R.; Zhu, Q.; Ma, S.; Wang, P.; Bi, X.; Zhang, X.; Yu, X.; Wu, Y.; Wu, Z. F.; Gou, Z.; Shao, Z.; Li, Z.; Gao, Z.; Liu, A.; Xue, B.; Wang, B.; Wu, B.; Feng, B.; Lu, C.; Zhao, C.; Deng, C.; Zhang, C.; Ruan, C.; Dai, D.; Chen, D.; Ji, D.; Li, E.; Lin, F.; Dai, F.; Luo, F.; Hao, G.; Chen, G.; Li, G.; Zhang, H.; Bao, H.; Xu, H.; Wang, H.; Ding, H.; Xin, H.; Gao, H.; Qu, H.; Li, H.; Guo, J.; Li, J.; Wang, J.; Chen, J.; Yuan, J.; Qiu, J.; Li, J.; Cai, J. L.; Ni, J.; Liang, J.; Chen, J.; Dong, K.; Hu, K.; Gao, K.; Guan, K.; Huang, K.; Yu, K.; Wang, L.; Zhang, L.; Zhao, L.; Wang, L.; Zhang, L.; Xu, L.; Xia, L.; Zhang, M.; Zhang, M.; Tang, M.; Li, M.; Wang, M.; Li, M.; Tian, N.; Huang, P.; Zhang, P.; Wang, Q.; Chen, Q.; Du, Q.; Ge, R.; Zhang, R.; Pan, R.; Wang, R.; Chen, R. J.; Jin, R. L.; Chen, R.; Lu, S.; Zhou, S.; Chen, S.; Ye, S.; Wang, S.; Yu, S.; Zhou, S.; Pan, S.; and Li, S. S. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *CoRR*, abs/2501.12948.
- Dubey, A.; Jauhri, A.; Pandey, A.; Kadian, A.; Al-Dahle, A.; Letman, A.; Mathur, A.; Schelten, A.; Yang, A.; Fan, A.; Goyal, A.; Hartshorn, A.; Yang, A.; Mitra, A.; Srivankumar, A.; Korenev, A.; Hinsvark, A.; Rao, A.; Zhang, A.; Rodriguez, A.; Gregerson, A.; Spataru, A.; Rozière, B.; Biron, B.; Tang, B.; Chern, B.; Caucheteux, C.; Nayak, C.; Bi, C.; Marra, C.; McConnell, C.; Keller, C.; Touret, C.; Wu, C.; Wong, C.; Ferrer, C. C.; Nikolaidis, C.; Allonsius, D.; Song, D.; Pintz, D.; Livshits, D.; Esiobu, D.; Choudhary, D.; Mahajan, D.; Garcia-Olano, D.; Perino, D.; Hupkes, D.; Lakomkin, E.; AlBadawy, E.; Lobanova, E.; Dinan, E.; Smith, E. M.; Radenovic, F.; Zhang, F.; Synnaeve, G.; Lee, G.; Anderson, G. L.; Nail, G.; Mialon, G.; Pang, G.; Cucurell, G.; Nguyen, H.; Korevaar, H.; Xu, H.; Touvron, H.; Zarov, I.; Ibarra, I. A.; Kloumann, I. M.; Misra, I.; Evtimov, I.; Copet, J.; Lee, J.; Geffert, J.; Vranes, J.; Park, J.; Mahadeokar, J.; Shah, J.; van der Linde, J.; Billock, J.; Hong, J.; Lee, J.; Fu, J.; Chi, J.; Huang, J.; Liu, J.; Wang, J.; Yu, J.; Bitton, J.; Spisak, J.; Park, J.; Rocca, J.; Johnstun, J.; Saxe, J.; Jia, J.; Alwala, K. V.; Upasani, K.; Plawiak, K.; Li, K.; Heafield, K.; Stone, K.; and et al. 2024. The Llama 3 Herd of Models. *CoRR*, abs/2407.21783.
- Guo, T.; Chen, X.; Wang, Y.; Chang, R.; Pei, S.; Chawla, N. V.; Wiest, O.; and Zhang, X. 2024. Large Language Model Based Multi-agents: A Survey of Progress and Challenges. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI 2024, Jeju, South Korea, August 3-9, 2024*, 8048–8057. ijcai.org.
- Jedidi, N.; Chuang, Y.; Glass, J. R.; and Lin, J. 2025. Don't "Overthink" Passage Reranking: Is Reasoning Truly Necessary? *CoRR*, abs/2505.16886.
- Ji, Y.; Li, Z.; Meng, R.; and He, D. 2024. ReasoningRank: Teaching Student Models to Rank through Reasoning-Based Knowledge Distillation. *CoRR*, abs/2410.05168.
- Li, X.; Shakir, A.; Huang, R.; Lipp, J.; and Li, J. 2025. ProRank: Prompt Warmup via Reinforcement Learning for Small Language Models Reranking. *CoRR*, abs/2506.03487.
- Luo, J.; Chen, X.; He, B.; and Sun, L. 2024. PRP-Graph: Pairwise Ranking Prompting to LLMs with Graph Aggregation for Effective Text Re-ranking. In Ku, L.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, Bangkok, Thailand, August 11-16, 2024, 5766–5776. Association for Computational Linguistics.
- Ma, X.; Wang, L.; Yang, N.; Wei, F.; and Lin, J. 2024. Fine-Tuning LLaMA for Multi-Stage Text Retrieval. In Yang, G. H.; Wang, H.; Han, S.; Hauff, C.; Zuccon, G.; and Zhang, Y., eds., *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, 2421–2425. ACM.
- Ma, X.; Zhang, X.; Pradeep, R.; and Lin, J. 2023. Zero-Shot Listwise Document Reranking with a Large Language Model. *CoRR*, abs/2305.02156.
- Nguyen, T.; Rosenberg, M.; Song, X.; Gao, J.; Tiwary, S.; Majumder, R.; and Deng, L. 2016. MS MARCO: A Human Generated MACHine Reading COMprehension Dataset. In Besold, T. R.; Bordes, A.; d'Avila Garcez, A. S.; and Wayne, G., eds., *Proceedings of the Workshop on Cognitive Computation: Integrating neural and symbolic approaches 2016 co-located with the 30th Annual Conference on Neural Information Processing Systems (NIPS 2016), Barcelona, Spain, December 9, 2016*, volume 1773 of *CEUR Workshop Proceedings*. CEUR-WS.org.
- Niu, T.; Joty, S.; Liu, Y.; Xiong, C.; Zhou, Y.; and Yavuz, S. 2024. JudgeRank: Leveraging Large Language Models for Reasoning-Intensive Reranking. *CoRR*, abs/2411.00142.
- Qin, Z.; Jagerman, R.; Hui, K.; Zhuang, H.; Wu, J.; Yan, L.; Shen, J.; Liu, T.; Liu, J.; Metzler, D.; Wang, X.; and Bendersky, M. 2024. Large Language Models are Effective Text Rankers with Pairwise Ranking Prompting. In Duh, K.; Gómez-Adorno, H.; and Bethard, S., eds., *Findings of the Association for Computational Linguistics: NAACL 2024, Mexico City, Mexico, June 16-21, 2024*, 1504–1518. Association for Computational Linguistics.

- Seetharaman, R.; Dhole, K. D.; and Bansal, A. 2025. InsertRank: LLMs can reason over BM25 scores to Improve Listwise Reranking. *CoRR*, abs/2506.14086.
- Shao, R.; Qiao, R.; Kishore, V.; Muennighoff, N.; Lin, X. V.; Rus, D.; Low, B. K. H.; Min, S.; Yih, W.; Koh, P. W.; and Zettlemoyer, L. 2025. ReasonIR: Training Retrievers for Reasoning Tasks. *CoRR*, abs/2504.20595.
- Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y.; et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Su, H.; Yen, H.; Xia, M.; Shi, W.; Muennighoff, N.; Wang, H.; Liu, H.; Shi, Q.; Siegel, Z. S.; Tang, M.; Sun, R.; Yoon, J.; Arik, S. Ö.; Chen, D.; and Yu, T. 2025. BRIGHT: A Realistic and Challenging Benchmark for Reasoning-Intensive Retrieval. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Sun, W.; Yan, L.; Ma, X.; Wang, S.; Ren, P.; Chen, Z.; Yin, D.; and Ren, Z. 2023. Is ChatGPT Good at Search? Investigating Large Language Models as Re-Ranking Agents. In Bouamor, H.; Pino, J.; and Bali, K., eds., *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023*, 14918–14937. Association for Computational Linguistics.
- Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models. *CoRR*, abs/2104.08663.
- Weller, O.; Chang, B.; MacAvaney, S.; Lo, K.; Cohan, A.; Durme, B. V.; Lawrie, D. J.; and Soldaini, L. 2025a. FollowIR: Evaluating and Teaching Information Retrieval Models to Follow Instructions. In Chiruzzo, L.; Ritter, A.; and Wang, L., eds., *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2025 - Volume 1: Long Papers, Albuquerque, New Mexico, USA, April 29 - May 4, 2025*, 11926–11942. Association for Computational Linguistics.
- Weller, O.; Ricci, K.; Yang, E.; Yates, A.; Lawrie, D.; and Van Durme, B. 2025b. Rank1: Test-time compute for reranking in information retrieval. *arXiv preprint arXiv:2502.18418*.
- Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; Zheng, C.; Liu, D.; Zhou, F.; Huang, F.; Hu, F.; Ge, H.; Wei, H.; Lin, H.; Tang, J.; Yang, J.; Tu, J.; Zhang, J.; Yang, J.; Yang, J.; Zhou, J.; Zhou, J.; Lin, J.; Dang, K.; Bao, K.; Yang, K.; Yu, L.; Deng, L.; Li, M.; Xue, M.; Li, M.; Zhang, P.; Wang, P.; Zhu, Q.; Men, R.; Gao, R.; Liu, S.; Luo, S.; Li, T.; Tang, T.; Yin, W.; Ren, X.; Wang, X.; Zhang, X.; Ren, X.; Fan, Y.; Su, Y.; Zhang, Y.; Zhang, Y.; Wan, Y.; Liu, Y.; Wang, Z.; Cui, Z.; Zhang, Z.; Zhou, Z.; and Qiu, Z. 2025a. Qwen3 Technical Report. *CoRR*, abs/2505.09388.
- Yang, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Li, C.; Liu, D.; Huang, F.; Wei, H.; Lin, H.; Yang, J.; Tu, J.; Zhang, J.; Yang, J.; Yang, J.; Zhou, J.; Lin, J.; Dang, K.; Bao, K.; Yang, K.; Yu, L.; Xue, M.; Zhang, P.; Men, R.; Lin, R.; Li, T.; Xia, T.; Ren, X.; Ren, X.; Fan, Y.; Su, Y.; Zhang, Y.; Wan, Y.; Liu, Y.; Cui, Z.; Zhang, Z.; and Qiu, Z. 2024. Qwen2.5 Technical Report. *CoRR*, abs/2412.15115.
- Yang, E.; Yates, A.; Ricci, K.; Weller, O.; Chari, V.; Durme, B. V.; and Lawrie, D. J. 2025b. Rank-K: Test-Time Reasoning for Listwise Reranking. *CoRR*, abs/2505.14432.
- Yu, Y.; Ping, W.; Liu, Z.; Wang, B.; You, J.; Zhang, C.; Shoenybi, M.; and Catanzaro, B. 2024. RankRAG: Unifying Context Ranking with Retrieval-Augmented Generation in LLMs. In Globersons, A.; Mackey, L.; Belgrave, D.; Fan, A.; Paquet, U.; Tomczak, J. M.; and Zhang, C., eds., *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Zhang, L.; Wang, B.; Qiu, X.; Reddy, S.; and Agrawal, A. 2025a. REARANK: Reasoning Re-ranking Agent via Reinforcement Learning. *arXiv preprint arXiv:2505.20046*.
- Zhang, Y.; Li, M.; Long, D.; Zhang, X.; Lin, H.; Yang, B.; Xie, P.; Yang, A.; Liu, D.; Lin, J.; Huang, F.; and Zhou, J. 2025b. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. *CoRR*, abs/2506.05176.
- Zhuang, S.; Ma, X.; Koopman, B.; Lin, J.; and Zuccon, G. 2025. Rank-r1: Enhancing reasoning in llm-based document rerankers via reinforcement learning. *arXiv preprint arXiv:2503.06034*.
- Zhuang, S.; Zhuang, H.; Koopman, B.; and Zuccon, G. 2024. A Setwise Approach for Effective and Highly Efficient Zero-shot Ranking with Large Language Models. In Yang, G. H.; Wang, H.; Han, S.; Hauff, C.; Zuccon, G.; and Zhang, Y., eds., *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, 38–47. ACM.

A Supplementary Material Overview

This appendix provides additional details and resources that complement the main paper. We include: (1) extended descriptions and statistics of the datasets used, (2) some comprehensive experimental results, and (3) further notes on inference settings and implementation.

B Dataset Supplementary Material

B.1 Dataset Statistics

Table B1 presents the detailed statistics of each training and evaluation data subset, including the number of queries, positive/negative pairs, and fine-grained annotations. Further breakdowns by task paradigm (pointwise, pairwise, listwise) are provided.

B.2 Prompt Formats

From Figure B1 to Figure B22, we provide the exact prompt templates and sample instances for each task paradigm (pointwise, pairwise, listwise), with or without fine-grained label, and different inference modes (`/think`, `/no think`).

C Experimental Supplementary Material

C.1 Full BEIR Results

Table C1 presents the complete results (NDCG@10) on specific BEIR benchmark datasets, which follows the setting of Rank1.

C.2 Significance Testing

To assess the robustness of our findings, we conduct paired significance tests (both paired t -test and Wilcoxon signed-rank test) on the per-query NDCG@10 scores for BRIGHT ($n = 1384$ queries). Table C2 reports p -values for multiple TFRank variants compared to REARANK-7B, under both BM25 and ReasonIR retrievers. Across all comparisons, both tests consistently yield highly significant p -values ($p \ll 0.01$), confirming that the improvements of TFRank over REARANK are statistically robust and not due to random variation.

D Additional Inference Notes

For all experiments, we use the vLLM inference framework, a fast and easy-to-use library for LLM inference and serving. During inference, the temperature is set to 0 to eliminate randomness and ensure deterministic outputs. The specific inference completion prompt template is shown in Figure B1.

Due to the current limitations of vLLM, only the top-20 tokens (by logit value) are accessible for extracting model scores (e.g., for yes, no, and the score tokens 0-4). If none of the target tokens appear in the top-20 results, we assign a neutral score of 0.5 to avoid errors. Notably, for all sufficiently trained TFRank models, such cases are sporadic.

Dataset	Source	Task	RR	FG	#Queries	#Samples
ms_sub_binary	MS MARCO, Rank1	Pointwise	N	N	195,335	385,791
ms_sub_finegrained	MS MARCO, M3 data, Deepseek-R1	Pointwise	Y	Y	7,068	$13,852 \times 2 \times 2$
		Pairwise	Y	Y	7,068	$57,351 \times 2 \times 2$
		Listwise	Y	Y	7,068	$44,605 \times 2 \times 2$

Table B1: Statistics of training data used in TFRank. “RR” indicates whether Relevance Reasoning (/think mode) is included (Y) or not (/no think mode, N); “FG” denotes the availability of fine-grained labels; “#” refers to the number of queries or samples. Multiplication terms (e.g., $\times 2 \times 2$) indicate duplication arising from both RR and FG settings.

Model	Approach	ArguA	ClimF	DBP	FiQA	NFCorp	SciDoc	SciFact	Touche	TrecC	Avg.	
BM25	-	Pointwise	39.7	16.5	31.8	23.6	33.8	14.9	67.9	44.2	59.5	36.9
RankGPT-4	Zero-shot	Listwise	-	-	47.1	-	38.5	-	75.0	38.6	85.5	-
Rank1-7B	SFT	Pointwise	26.4	16.2	37.7	38.4	37.9	16.5	76.1	24.5	79.5	39.2
Rank1-14B	SFT	Pointwise	32.2	15.6	34.2	36.6	35.1	16.6	73.8	25.9	78.0	38.7
REARANK-7B	GRPO	Listwise	35.6	20.6	43.5	35.8	37.9	19.2	71.9	40.2	80.1	42.8
Rank-R1-3B	GRPO	Setwise	31.5	22.5	41.2	29.8	31.8	16.3	64.0	27.4	75.9	37.8
Rank-R1-7B	GRPO	Setwise	37.0	24.1	43.2	40.1	36.2	18.8	76.1	33.0	82.6	43.5
Rank-R1-14B	GRPO	Setwise	34.4	24.2	44.0	43.0	37.9	19.7	77.5	29.6	83.9	43.8
Llama3.2 Series												
TFRank-1B	SFT	Pointwise	27.4	18.5	35.4	19.5	23.6	9.7	47.3	22.6	65.4	29.9
TFRank-3B	SFT	Pointwise	34.2	16.6	38.1	33.3	36.2	15.8	72.3	35.1	82.5	40.5
TFRank-8B	SFT	Pointwise	38.5	22.7	42.6	38.0	37.5	18.3	75.6	33.0	82.5	43.2
Qwen2.5 Series												
TFRank-0.5B	SFT	Pointwise	29.7	17.8	37.0	28.3	33.6	13.2	65.1	25.6	78.7	36.6
TFRank-1.5B	SFT	Pointwise	36.5	17.8	38.1	31.4	35.1	15.5	71.8	32.2	77.4	39.5
TFRank-3B	SFT	Pointwise	33.2	18.8	38.3	33.2	36.0	15.6	71.8	32.0	78.0	39.7
TFRank-7B	SFT	Pointwise	35.7	20.0	38.5	34.8	37.6	19.1	75.0	33.0	80.0	41.5
Qwen3 Series												
TFRank-0.6B	SFT	Pointwise	35.4	21.8	34.3	30.5	32.8	14.3	71.8	29.8	80.7	39.0
TFRank-1.7B	SFT	Pointwise	37.0	21.3	38.8	31.1	33.2	16.1	70.9	29.3	76.8	39.4
TFRank-4B	SFT	Pointwise	37.2	19.7	37.9	36.2	38.3	18.3	76.6	37.6	80.5	42.5
TFRank-8B	SFT	Pointwise	36.5	21.7	37.0	38.0	38.0	17.9	74.6	35.0	80.0	42.1

Table C1: Main evaluation results (NDCG@10) on the BEIR benchmark using BM25 as retriever. TFRank models are trained on their respective backbones. For each column, the highest value is **bolded** and the second highest is underlined.

Retriever	Ranker Comparison	T-test	Wilcoxon
BM25 (pyserini)	TFRank-7B SFT (Qwen2.5) vs. REARANK-7B	1.47×10^{-3}	5.70×10^{-4}
BM25 (pyserini)	TFRank-8B SFT (Qwen3) vs. REARANK-7B	1.47×10^{-12}	4.14×10^{-13}
BM25 (pyserini)	TFRank-8B SFT+GRPO (Qwen3) vs. REARANK-7B	5.70×10^{-17}	8.46×10^{-18}
ReasonIR-8B	TFRank-8B SFT+GRPO (Qwen3) vs. REARANK-7B	4.57×10^{-14}	1.01×10^{-14}

Table C2: Some paired significance tests (p -values) for NDCG@10 on BRIGHT ($n = 1384$ queries). Each row compares TFRank and REARANK-7B under the specified retriever and ranker configuration. Both paired t -test and Wilcoxon signed-rank test are reported.

(a) /think

```
<im_start>system
Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.
<im_end>
<im_start>user
<Instruct>:
<Query>:
<Document>:
/think
<im_end>
<im_start>assistant
```

(b) /no think

```
<im_start>system
Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.
<im_end>
<im_start>user
<Instruct>:
<Query>:
<Document>:
/no think
<im_end>
<im_start>assistant
<think>

</think>
```

Figure B1: Prompt templates for TFRank inference. (a) depicts the completion prompt for the `/think` mode, which requires the model to output an explicit reasoning process. (b) shows the template for the `/no think` mode, instructing the model to return a relevance score without reasoning directly.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please judge the relevance strength between the query and the document, and directly output the relevance judgment as yes or no.

<Query>: what is a stereo preamplifier

<Document>: Amplifiers are essential components in any sound system, boosting the audio signal to drive loudspeakers and produce audible sound. /think

[assistant] <think>

To evaluate relevance, I first analyze the query's core intent: it seeks a definition or explanation of a 'stereo preamplifier' – a specialized component that conditions audio signals before amplification. The document discusses amplifiers generically, explaining their role in boosting signals for speakers. While amplifiers and preamplifiers exist in the same audio ecosystem, they serve distinct functions: preamplifiers handle input selection and signal preparation, whereas amplifiers focus on power output. The document mentions neither 'preamplifier' nor its unique functions (e.g., impedance matching, phono equalization). It only addresses the amplification stage, omitting the pre-amplification context critical to the query. This creates a tangential relationship – the content relates broadly to audio hardware but fails to address the specific device inquired about. Relevance levels exist on a spectrum: 'Completely Irrelevant' implies zero conceptual overlap; 'Weakly Relevant' indicates peripheral connections without substantive alignment; 'Moderately Relevant' suggests partial topic coverage; 'Strongly Relevant' denotes direct addressing of core aspects; 'Completely Relevant' requires exhaustive query fulfillment. Here, the document's focus on amplification principles provides background context tangentially linked to audio systems but lacks any direct engagement with preamplifier functionality or definition. The connection remains superficial and indirect.

The document exhibits low relevance to the query. While it accurately describes the general function of amplifiers in audio systems – boosting signals to drive speakers – it does not address the specific subject of 'stereo preamplifiers.' Preamplifiers serve a distinct purpose in signal conditioning prior to amplification, a concept entirely absent from the document. The content provides only peripheral context about audio hardware without engaging with the query's core request for information about preamplifiers. Consequently, the document is weakly related to the user's information need, aligning with the reference assessment of 'weakly related'.

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, the final relevance judgment should be: no. I will directly output the judgment according to the required format.

</think>

no

Figure B2: Example of a pointwise binary (yes/no) training sample in messages format with explicit reasoning (/think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please judge the relevance strength between the query and the document, and directly output the relevance judgment as yes or no.

<Query>: what is a stereo preamplifier

<Document>: Amplifiers are essential components in any sound system, boosting the audio signal to drive loudspeakers and produce audible sound. /no think

[assistant] <think>

</think>

no

Figure B3: Example of a pointwise binary (yes/no) training sample in messages format without explicit reasoning (/no think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please judge the relevance strength between the query and the document, and directly output the relevance judgment (yes or no), followed by the relevance score in parentheses, e.g., yes(score) or no(score).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:
0 means completely irrelevant,
1 means weakly relevant,
2 means moderately relevant,
3 means strongly relevant,
4 means completely relevant.

- For binary relevance judgment (yes or no), the rule is:
Scores 0 and 1 are considered irrelevant and represented as "no",
Scores 2, 3, and 4 are considered relevant and represented as "yes".

<Query>: what is a stereo preamplifier

<Document>: Amplifiers are essential components in any sound system, boosting the audio signal to drive loudspeakers and produce audible sound. /think

[assistant] <think>

To evaluate relevance, I first analyze the query's core intent: it seeks a definition or explanation of a 'stereo preamplifier' – a specialized component that conditions audio signals before amplification. The document discusses amplifiers generically, explaining their role in boosting signals for speakers. While amplifiers and preamplifiers exist in the same audio ecosystem, they serve distinct functions: preamplifiers handle input selection and signal preparation, whereas amplifiers focus on power output. The document mentions neither 'preamplifier' nor its unique functions (e.g., impedance matching, phono equalization). It only addresses the amplification stage, omitting the pre-amplification context critical to the query. This creates a tangential relationship – the content relates broadly to audio hardware but fails to address the specific device inquired about. Relevance levels exist on a spectrum: 'Completely Irrelevant' implies zero conceptual overlap; 'Weakly Relevant' indicates peripheral connections without substantive alignment; 'Moderately Relevant' suggests partial topic coverage; 'Strongly Relevant' denotes direct addressing of core aspects; 'Completely Relevant' requires exhaustive query fulfillment. Here, the document's focus on amplification principles provides background context tangentially linked to audio systems but lacks any direct engagement with preamplifier functionality or definition. The connection remains superficial and indirect.

The document exhibits low relevance to the query. While it accurately describes the general function of amplifiers in audio systems – boosting signals to drive speakers – it does not address the specific subject of 'stereo preamplifiers.' Preamplifiers serve a distinct purpose in signal conditioning prior to amplification, a concept entirely absent from the document. The content provides only peripheral context about audio hardware without engaging with the query's core request for information about preamplifiers. Consequently, the document is weakly related to the user's information need, aligning with the reference assessment of 'weakly related'.

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, the final relevance score should be: 1, combined with the relevance score explanation, the relevance judgment should be no. I will output the judgment with the score according to the required format.

</think>

no(1)

Figure B4: Example of a pointwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please judge the relevance strength between the query and the document, and directly output the relevance judgment (yes or no), followed by the relevance score in parentheses, e.g., yes(score) or no(score).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:
0 means completely irrelevant,
1 means weakly relevant,
2 means moderately relevant,
3 means strongly relevant,
4 means completely relevant.

- For binary relevance judgment (yes or no), the rule is:
Scores 0 and 1 are considered irrelevant and represented as "no",
Scores 2, 3, and 4 are considered relevant and represented as "yes".

<Query>: what is a stereo preamplifier

<Document>: Amplifiers are essential components in any sound system, boosting the audio signal to drive loudspeakers and produce audible sound. /no think

[assistant] <think>

</think>

no(1)

Figure B5: Example of a pointwise fine-grained training sample in messages format without explicit reasoning (/no think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please compare the relevance of two documents based on the query, and determine which document is more relevant to the query, or if both have the same relevance.

Instructions:

- Directly output the comparison expression,
- Use ">" and "=" to connect document indices to indicate relevance relationships,

<Query>: what is a stereo preamplifier

<Documents>:

[1] A preamplifier takes the delicate, low-voltage signals from sources like phono cartridges or line-level inputs and prepares them for the more robust amplification stage.

[2] Marshall Amplification Marshall Amplification. Marshall Amplification is an English company that designs and manufactures music amplifiers, speaker cabinets, brands personal headphones and earphones, and, having acquired Natal Drums, drums and bongos. It was founded by drum shop owner and drummer Jim Marshall, and is now based in Bletchley, Milton Keynes, Buckinghamshire. Marshall's guitar amplifiers are among the most recognised in the world. They are known for their Marshall crunch. This signature sound was conceived by Marshall after gu

/think

[assistant] <think>

To evaluate pairwise relevance, I first consider the core concept of 'relevance' in information retrieval. Relevance fundamentally measures how effectively a document satisfies the information need expressed in a query. This involves examining both topical alignment (does the document address the query subject?) and functional utility (does it provide explanatory value for the query?). For 'what is a stereo preamplifier', the ideal document should define preamplifiers, explain their function in stereo systems, and differentiate them from other components.

Analyzing Document [1]: It explicitly defines a preamplifier's core function - processing low-voltage signals from input sources to prepare them for amplification. While it doesn't specify 'stereo', the described functionality inherently applies to stereo systems. It covers essential aspects: signal handling (delicate low-voltage signals), input sources (phono cartridges, line-level), and purpose (preparation for main amplification). This provides direct explanatory value for understanding preamplifiers.

Figure B6: Example of a pairwise ID-only training sample in messages format with explicit reasoning (/think mode) in TFRank (1/2).

Analyzing Document [2]: This focuses exclusively on Marshall Amplification as a company - its history, product range (guitar amplifiers, cabinets, headphones, drums), and brand identity ('Marshall crunch'). There's no mention of preamplifiers, signal processing, or component definitions. The content revolves around corporate biography and guitar-specific amplification, which is unrelated to the query's request for a technical explanation of stereo preamplifiers.

Comparing the documents: [1] demonstrates moderate relevance by directly addressing the core concept of a preamplifier's function, which is fundamental to understanding 'what' it is. [2] is completely unrelated as it discusses a guitar amplifier manufacturer without any connection to preamplifier functionality or definitions. The distinction is clear: [1] provides explanatory content aligned with the query's intent, while [2] discusses an entirely different product category and company history. Therefore, the pairwise ranking is unequivocal: [1] > [2].

Document [1] is moderately relevant as it directly explains the core function of a preamplifier—processing low-level signals for subsequent amplification—which fundamentally addresses the query's request for an explanation of what a stereo preamplifier is. Document [2] is completely unrelated, focusing solely on a guitar amplifier manufacturer's history and products without any mention of preamplifiers or their functionality. Consequently, Document [1] is significantly more relevant than Document [2] ([1] > [2]).

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, the final comparison result should be: [1] > [2]. I will output the comparison expression according to the required format.

</think>

[1] > [2]

Figure B7: Example of a pairwise ID-only training sample in `messages` format with explicit reasoning (`/think` mode) in TFRank (2/2).

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please compare the relevance of two documents based on the query, and determine which document is more relevant to the query, or if both have the same relevance.

Instructions:

- Directly output the comparison expression,
- Use ">" and "=" to connect document indices to indicate relevance relationships,

<Query>: what is a stereo preamplifier

<Documents>:

[1] A preamplifier takes the delicate, low-voltage signals from sources like phono cartridges or line-level inputs and prepares them for the more robust amplification stage.

[2] Marshall Amplification Marshall Amplification. Marshall Amplification is an English company that designs and manufactures music amplifiers, speaker cabinets, brands personal headphones and earphones, and, having acquired Natal Drums, drums and bongos. It was founded by drum shop owner and drummer Jim Marshall, and is now based in Bletchley, Milton Keynes, Buckinghamshire. Marshall's guitar amplifiers are among the most recognised in the world. They are known for their Marshall crunch. This signature sound was conceived by Marshall after gu

/no think

[assistant] <think>

</think>

[1] > [2]

Figure B8: Example of a pairwise ID-only training sample in `messages` format without explicit reasoning (`/no think` mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please compare the relevance of two documents based on the query, and determine which document is more relevant to the query, or if both have the same relevance.

Instructions:

- Directly output the comparison expression,
- Use ">" and "=" to connect document indices to indicate relevance relationships,
- In the comparison expression, annotate each document with its relevance score based on the relevance score description, in the format [DocumentIndex](RelevanceScore).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:

- 0 means completely irrelevant,
- 1 means weakly relevant,
- 2 means moderately relevant,
- 3 means strongly relevant,
- 4 means completely relevant.

<Query>: what is a stereo preamplifier

<Documents>:

[1] A preamplifier takes the delicate, low-voltage signals from sources like phono cartridges or line-level inputs and prepares them for the more robust amplification stage.

[2] Marshall Amplification Marshall Amplification. Marshall Amplification is an English company that designs and manufactures music amplifiers, speaker cabinets, brands personal headphones and earphones, and, having acquired Natal Drums, drums and bongos. It was founded by drum shop owner and drummer Jim Marshall, and is now based in Bletchley, Milton Keynes, Buckinghamshire. Marshall's guitar amplifiers are among the most recognised in the world. They are known for their Marshall crunch. This signature sound was conceived by Marshall after gu

/think

[assistant] <think>

To evaluate pairwise relevance, I first consider the core concept of 'relevance' in information retrieval. Relevance fundamentally measures how effectively a document satisfies the information need expressed in a query. This involves examining both topical alignment (does the document address the query subject?) and functional utility (does it provide explanatory value for the query?). For 'what is a stereo preamplifier', the ideal document should define preamplifiers, explain their function in stereo systems, and differentiate them from other components.

Figure B9: Example of a pairwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (1/2).

Analyzing Document [1]: It explicitly defines a preamplifier's core function - processing low-voltage signals from input sources to prepare them for amplification. While it doesn't specify 'stereo', the described functionality inherently applies to stereo systems. It covers essential aspects: signal handling (delicate low-voltage signals), input sources (phono cartridges, line-level), and purpose (preparation for main amplification). This provides direct explanatory value for understanding preamplifiers.

Analyzing Document [2]: This focuses exclusively on Marshall Amplification as a company - its history, product range (guitar amplifiers, cabinets, headphones, drums), and brand identity ('Marshall crunch'). There's no mention of preamplifiers, signal processing, or component definitions. The content revolves around corporate biography and guitar-specific amplification, which is unrelated to the query's request for a technical explanation of stereo preamplifiers.

Comparing the documents: [1] demonstrates moderate relevance by directly addressing the core concept of a preamplifier's function, which is fundamental to understanding 'what' it is. [2] is completely unrelated as it discusses a guitar amplifier manufacturer without any connection to preamplifier functionality or definitions. The distinction is clear: [1] provides explanatory content aligned with the query's intent, while [2] discusses an entirely different product category and company history. Therefore, the pairwise ranking is unequivocal: [1] > [2].

Document [1] is moderately relevant as it directly explains the core function of a preamplifier—processing low-level signals for subsequent amplification—which fundamentally addresses the query's request for an explanation of what a stereo preamplifier is. Document [2] is completely unrelated, focusing solely on a guitar amplifier manufacturer's history and products without any mention of preamplifiers or their functionality. Consequently, Document [1] is significantly more relevant than Document [2] ([1] > [2]).

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, Based on relevance scores, Document [1] has a score of 2, Document [2] has a score of 0. the final comparison result should be: [1] > [2]. I will output the comparison expression with scores according to the required format.

</think>

[1](2) > [2](0)

Figure B10: Example of a pairwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (2/2).

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please compare the relevance of two documents based on the query, and determine which document is more relevant to the query, or if both have the same relevance.

Instructions:

- Directly output the comparison expression,
- Use ">" and "=" to connect document indices to indicate relevance relationships,
- In the comparison expression, annotate each document with its relevance score based on the relevance score description, in the format [DocumentIndex](RelevanceScore).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:
0 means completely irrelevant,
1 means weakly relevant,
2 means moderately relevant,
3 means strongly relevant,
4 means completely relevant.

<Query>: what is a stereo preamplifier

<Documents>:

[1] A preamplifier takes the delicate, low-voltage signals from sources like phono cartridges or line-level inputs and prepares them for the more robust amplification stage.

[2] Marshall Amplification Marshall Amplification. Marshall Amplification is an English company that designs and manufactures music amplifiers, speaker cabinets, brands personal headphones and earphones, and, having acquired Natal Drums, drums and bongos. It was founded by drum shop owner and drummer Jim Marshall, and is now based in Bletchley, Milton Keynes, Buckinghamshire. Marshall's guitar amplifiers are among the most recognised in the world. They are known for their Marshall crunch. This signature sound was conceived by Marshall after gu
/no think

[assistant] <think>

</think>

[1](2) > [2](0)

Figure B11: Example of a pairwise fine-grained training sample in messages format without explicit reasoning (/no think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please rank the list of documents based on their relevance to the query.

Instructions:

- Directly output the list ranking expression,
- Documents with higher relevance should be placed in front,
- Use ">" and "=" to connect document indices to indicate relevance relationships,

<Query>: fisher price musical play mat

<Documents>:

[1] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[2] - The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.

[3] A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.

[4] Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.

[5] Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.

[6] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[7] Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.

[8] Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.

[9] Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.

[10] How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.

/think

[assistant] <think>

The task at hand involves assessing the degree of connection between a user's information need, expressed as a query, and various textual documents. This process, known as relevance ranking, requires a nuanced understanding of 'relevance' itself.

****Understanding Relevance: An Abstract Perspective****

At its core, 'relevance' describes the degree to which information in a document is pertinent, applicable, or helpful in addressing the user's query or underlying intent. It's not a binary concept (relevant/irrelevant) but a spectrum, ranging from absolute non-connection to perfect alignment. This spectrum can be delineated into several distinct levels:

* ****Completely Irrelevant (No Connection):**** Documents in this category bear no discernible relationship to the query. They discuss entirely different subjects, contain no matching keywords in a meaningful context, and offer no utility whatsoever to the user's information seeking. Their presence is pure noise.

Figure B12: Example of a listwise ID-only training sample in messages format with explicit reasoning (/think mode) in TFRank (1/4).

* **Weakly Relevant (Slightly Related):** These documents share a very loose connection, perhaps mentioning one or two keywords from the query, but the overall topic or specific product type is fundamentally different. They might be tangentially related by a brand name or a very broad category, but the core intent of the query is not addressed.

* **Moderately Relevant (Somewhat Related):** Documents at this level hit a significant portion of the query's components but miss a crucial one, or they address the query's general subject but lack the specificity requested. They offer partial utility but require the user to bridge gaps or overlook missing details.

* **Strongly Relevant (Highly Relevant):** These documents are very closely aligned with the query's intent and keywords. They might not be an exact match in every single aspect, but they provide substantial, direct, and useful information about the queried topic or product. They clearly pertain to the user's need.

* **Completely Relevant (Perfectly Matched):** This represents the zenith of relevance. The document fully and precisely addresses the query, containing all the key terms in the correct context, and directly fulfills the user's information need without any ambiguity or significant missing information. It is essentially what the user was looking for.

In a listwise ranking scenario, the process isn't just about evaluating each document in isolation but also comparing them against each other to establish a coherent order of utility.

Query Analysis:

The query is 'fisher price musical play mat'. This query has three core components:

1. **Brand:** 'Fisher-Price'
2. **Attribute/Feature:** 'musical'
3. **Product Type:** 'play mat'

The user's intent is clearly to find a specific product: a play mat made by the Fisher-Price brand that incorporates musical features. Any document that addresses all three components accurately will be highly relevant. Documents that address only some components, or relate to similar but not identical products, will exhibit varying degrees of lower relevance.

Document-Specific Relevance Assessment and Comparison:

Let's meticulously analyze each document against the query and against each other:

* **Document [1]:** 'Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.'

* **Analysis:** This document is a set of search results that specifically list multiple products: 'Fisher-Price Kick and Play Piano Gym', 'Fisher-Price Music and Lights Deluxe Gym', and 'Fisher-Price 3-in-1 Musical Activity Gym'. All these explicitly contain 'Fisher-Price', imply 'play mat' (gym/mat context), and prominently feature 'musical' ('Piano Gym', 'Music and Lights', 'Musical Activity'). It directly provides examples of the exact product sought.

* **Relevance Level:** **Completely Relevant**. It's an ideal match, fulfilling all aspects of the query's intent by showing specific products.

* **Document [2]:** '- The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.'

* **Analysis:** This document is an academic legal article. It contains no keywords or conceptual links to 'Fisher-Price', 'musical', or 'play mat'.

* **Relevance Level:** **Completely Irrelevant**. There is absolutely no connection.

Figure B13: Example of a listwise ID-only training sample in messages format with explicit reasoning (/think mode) in TFRank (2/4).

* **Document [3]:** 'A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.'

* **Analysis:** This document is a guide on 'musical baby activity gym' which is a strong synonym for 'musical play mat' (product type + musical attribute). It mentions 'mats with piano keys or sound effects', reinforcing the 'musical' aspect. However, it does not mention 'Fisher-Price' specifically, discussing 'many brands'.

* **Relevance Level:** **Moderately Relevant**. It perfectly captures the 'musical play mat' aspect, but misses the brand specificity. It offers general information highly relevant to the product category.

* **Document [4]:** 'Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.'

* **Analysis:** This document mentions 'Fisher-Price' (brand) and 'Activity Mat' (a type of play mat). However, it completely lacks any mention or implication of 'musical' features. It describes a generic Fisher-Price play mat.

* **Relevance Level:** **Moderately Relevant**. It hits two out of three key components (brand and product type), but critically misses the 'musical' attribute. It's a Fisher-Price play mat, but not *the* musical one.

* **Document [5]:** 'Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.'

* **Analysis:** This document specifically discusses 'Fisher-Price Deluxe Kick & Play Piano Gym'. The 'Piano Gym' clearly implies a 'musical play mat', and the text explicitly mentions 'musical features' and 'mat design'. It's a review *of* the exact type of product the user is looking for.

* **Relevance Level:** **Strongly Relevant**. It pertains directly to a specific product that fully matches the query's criteria ('Fisher-Price musical play mat'), providing valuable insights (reviews). It's not a direct product listing, but it's about *the* product.

* **Document [6]:** 'Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.'

* **Analysis:** This document is identical to Document [1].

* **Relevance Level:** **Completely Relevant**. Same reasoning as Document [1].

* **Document [7]:** 'Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.'

* **Analysis:** This document mentions 'Fisher-Price' (brand) and 'songs and phrases' (implying musical/audio features). However, the product types are 'Train' and 'Laptop', which are explicitly *not* 'play mats'.

* **Relevance Level:** **Weakly Relevant**. It connects on brand and musicality, but fails on the fundamental product type. The mismatch on product type makes it only slightly related.

* **Document [8]:** 'Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.'

* **Analysis:** This document only mentions 'Fisher-Price' (brand). It's a 'Rock-a-Stack' which is neither a 'play mat' nor 'musical'.

* **Relevance Level:** **Weakly Relevant**. The connection is extremely tenuous, limited to only the brand name. It's a Fisher-Price toy, but completely unrelated to 'musical' or 'play mat'.

* **Document [9]:** 'Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.'

* **Analysis:** This document is a food recipe. It contains no keywords or conceptual links to the query.

* **Relevance Level:** **Completely Irrelevant**. No connection whatsoever.

Figure B14: Example of a listwise ID-only training sample in messages format with explicit reasoning (/think mode) in TFRank (3/4).

* **Document [10]:** 'How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.'

* **Analysis:** This document explicitly mentions 'Fisher-Price musical baby play mat' and is a guide on how to clean it. It directly refers to the exact product type and brand specified in the query.

* **Relevance Level:** **Strongly Relevant**. While not a product listing, it is highly pertinent as it provides practical information about the very product the user is searching for. It confirms the existence and offers utility regarding the specific item.

Synthesizing the Ranking:

Based on the detailed analysis and comparisons, the documents naturally fall into distinct tiers of relevance:

1. **Top Tier (Completely Relevant):** Documents [1] and [6] are identical and provide direct product listings for Fisher-Price musical play mats. They offer immediate and precise fulfillment of the query.
2. **Second Tier (Strongly Relevant):** Documents [5] and [10] discuss the specific Fisher-Price musical play mat. Document [5] provides user reviews, and Document [10] offers cleaning instructions. Both are highly useful and directly pertain to the existence and characteristics of the queried product.
3. **Third Tier (Moderately Relevant):** Documents [3] and [4] partially meet the query's criteria. Document [3] is a general guide to musical play mats but lacks brand specificity. Document [4] is a Fisher-Price play mat but lacks the musical feature. They offer partial matches but miss a key component.
4. **Fourth Tier (Weakly Relevant):** Documents [7] and [8] have very limited connections. Document [7] is a Fisher-Price musical toy but not a play mat. Document [8] is merely a Fisher-Price toy, with no musical or play mat relevance.
5. **Bottom Tier (Completely Irrelevant):** Documents [2] and [9] have no relevance at all to the query's topic.

This logical progression forms the basis for the final ranking.

Here is a comprehensive assessment of the relevance of each document to the query 'fisher price musical play mat', categorized by their connection to the user's specific information need:

- * Document [1] demonstrates a complete alignment with the query's intent, presenting exact matches for products like 'Fisher-Price Kick and Play Piano Gym' and 'Fisher-Price Music and Lights Deluxe Gym'.
- * Document [2] bears no connection whatsoever to the query's subject matter.
- * Document [3] is moderately related, offering a guide on 'musical baby activity gyms' which is a direct product category, yet it omits the specific 'Fisher-Price' brand mention.
- * Document [4] is also moderately related, featuring a 'Fisher-Price Activity Mat', but crucially lacks any musical attribute.
- * Document [5] is closely related, providing customer reviews specifically for a 'Fisher-Price Deluxe Kick & Play Piano Gym', highlighting its musical features.
- * Document [6] is completely relevant, being an identical set of search results to Document [1], directly fulfilling the query with specific musical play mats by Fisher-Price.
- * Document [7] is somewhat related, as it mentions 'Fisher-Price' and 'songs and phrases', but refers to products like a 'laptop' or 'train', not a 'play mat'.
- * Document [8] is weakly related, mentioning only 'Fisher-Price' and a generic toy ('Rock-a-Stack'), with no musical or play mat relevance.
- * Document [9] is entirely unrelated, discussing a recipe for lasagna.
- * Document [10] is significantly relevant, providing cleaning tips specifically for a 'Fisher-Price musical baby play mat', directly addressing the queried item.

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, the final document ranking should be: [6] = [1] > [10] = [5] > [3] = [4] > [8] = [7] > [9] = [2]. I will output the ranking expression according to the required format.

</think>

[6] = [1] > [10] = [5] > [3] = [4] > [8] = [7] > [9] = [2]

Figure B15: Example of a listwise ID-only training sample in messages format with explicit reasoning (/think mode) in TFRank (4/4).

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please rank the list of documents based on their relevance to the query.

Instructions:

- Directly output the list ranking expression,
- Documents with higher relevance should be placed in front,
- Use ">" and "=" to connect document indices to indicate relevance relationships,

<Query>: fisher price musical play mat

<Documents>:

[1] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[2] - The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.

[3] A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.

[4] Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.

[5] Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.

[6] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[7] Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.

[8] Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.

[9] Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.

[10] How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.

/no think

[assistant] <think>

</think>

[6] = [1] > [10] = [5] > [3] = [4] > [8] = [7] > [9] = [2]

Figure B16: Example of a listwise ID-only training sample in messages format without explicit reasoning (/no think mode) in TFRank.

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please rank the list of documents based on their relevance to the query.

Instructions:

- Directly output the list ranking expression,
- Documents with higher relevance should be placed in front,
- Use ">" and "=" to connect document indices to indicate relevance relationships,
- In the list ranking expression, annotate each document with its relevance score based on the relevance score description, in the format [DocumentIndex](RelevanceScore).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:

0 means completely irrelevant,

1 means weakly relevant,

2 means moderately relevant,

3 means strongly relevant,

4 means completely relevant.

<Query>: fisher price musical play mat

<Documents>:

[1] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[2] - The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.

[3] A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.

[4] Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.

[5] Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.

[6] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[7] Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.

[8] Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.

[9] Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.

[10] How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.

/think

[assistant] <think>

The task at hand involves assessing the degree of connection between a user's information need, expressed as a query, and various textual documents. This process, known as relevance ranking, requires a nuanced understanding of 'relevance' itself.

****Understanding Relevance: An Abstract Perspective****

Figure B17: Example of a listwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (1/5).

At its core, 'relevance' describes the degree to which information in a document is pertinent, applicable, or helpful in addressing the user's query or underlying intent. It's not a binary concept (relevant/irrelevant) but a spectrum, ranging from absolute non-connection to perfect alignment. This spectrum can be delineated into several distinct levels:

* **Completely Irrelevant (No Connection):** Documents in this category bear no discernible relationship to the query. They discuss entirely different subjects, contain no matching keywords in a meaningful context, and offer no utility whatsoever to the user's information seeking. Their presence is pure noise.

* **Weakly Relevant (Slightly Related):** These documents share a very loose connection, perhaps mentioning one or two keywords from the query, but the overall topic or specific product type is fundamentally different. They might be tangentially related by a brand name or a very broad category, but the core intent of the query is not addressed.

* **Moderately Relevant (Somewhat Related):** Documents at this level hit a significant portion of the query's components but miss a crucial one, or they address the query's general subject but lack the specificity requested. They offer partial utility but require the user to bridge gaps or overlook missing details.

* **Strongly Relevant (Highly Relevant):** These documents are very closely aligned with the query's intent and keywords. They might not be an exact match in every single aspect, but they provide substantial, direct, and useful information about the queried topic or product. They clearly pertain to the user's need.

* **Completely Relevant (Perfectly Matched):** This represents the zenith of relevance. The document fully and precisely addresses the query, containing all the key terms in the correct context, and directly fulfills the user's information need without any ambiguity or significant missing information. It is essentially what the user was looking for.

In a listwise ranking scenario, the process isn't just about evaluating each document in isolation but also comparing them against each other to establish a coherent order of utility.

Query Analysis:

The query is 'fisher price musical play mat'. This query has three core components:

1. **Brand:** 'Fisher-Price'
2. **Attribute/Feature:** 'musical'
3. **Product Type:** 'play mat'

The user's intent is clearly to find a specific product: a play mat made by the Fisher-Price brand that incorporates musical features. Any document that addresses all three components accurately will be highly relevant. Documents that address only some components, or relate to similar but not identical products, will exhibit varying degrees of lower relevance.

Document-Specific Relevance Assessment and Comparison:

Let's meticulously analyze each document against the query and against each other:

* **Document [1]:** 'Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.'

* **Analysis:** This document is a set of search results that specifically list multiple products: 'Fisher-Price Kick and Play Piano Gym', 'Fisher-Price Music and Lights Deluxe Gym', and 'Fisher-Price 3-in-1 Musical Activity Gym'. All these explicitly contain 'Fisher-Price', imply 'play mat' (gym/mat context), and prominently feature 'musical' ('Piano Gym', 'Music and Lights', 'Musical Activity'). It directly provides examples of the exact product sought.

* **Relevance Level:** **Completely Relevant**. It's an ideal match, fulfilling all aspects of the query's intent by showing specific products.

Figure B18: Example of a listwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (2/5).

* **Document [2]:** '- The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.'

* **Analysis:** This document is an academic legal article. It contains no keywords or conceptual links to 'Fisher-Price', 'musical', or 'play mat'.

* **Relevance Level:** **Completely Irrelevant**. There is absolutely no connection.

* **Document [3]:** 'A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.'

* **Analysis:** This document is a guide on 'musical baby activity gym' which is a strong synonym for 'musical play mat' (product type + musical attribute). It mentions 'mats with piano keys or sound effects', reinforcing the 'musical' aspect. However, it does not mention 'Fisher-Price' specifically, discussing 'many brands'.

* **Relevance Level:** **Moderately Relevant**. It perfectly captures the 'musical play mat' aspect, but misses the brand specificity. It offers general information highly relevant to the product category.

* **Document [4]:** 'Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.'

* **Analysis:** This document mentions 'Fisher-Price' (brand) and 'Activity Mat' (a type of play mat). However, it completely lacks any mention or implication of 'musical' features. It describes a generic Fisher-Price play mat.

* **Relevance Level:** **Moderately Relevant**. It hits two out of three key components (brand and product type), but critically misses the 'musical' attribute. It's a Fisher-Price play mat, but not *the* musical one.

* **Document [5]:** 'Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.'

* **Analysis:** This document specifically discusses 'Fisher-Price Deluxe Kick & Play Piano Gym'. The 'Piano Gym' clearly implies a 'musical play mat', and the text explicitly mentions 'musical features' and 'mat design'. It's a review *of* the exact type of product the user is looking for.

* **Relevance Level:** **Strongly Relevant**. It pertains directly to a specific product that fully matches the query's criteria ('Fisher-Price musical play mat'), providing valuable insights (reviews). It's not a direct product listing, but it's about *the* product.

* **Document [6]:** 'Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.'

* **Analysis:** This document is identical to Document [1].

* **Relevance Level:** **Completely Relevant**. Same reasoning as Document [1].

* **Document [7]:** 'Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.'

* **Analysis:** This document mentions 'Fisher-Price' (brand) and 'songs and phrases' (implying musical/audio features). However, the product types are 'Train' and 'Laptop', which are explicitly *not* 'play mats'.

* **Relevance Level:** **Weakly Relevant**. It connects on brand and musicality, but fails on the fundamental product type. The mismatch on product type makes it only slightly related.

* **Document [8]:** 'Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.'

Figure B19: Example of a listwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (3/5).

* **Analysis:** This document only mentions 'Fisher-Price' (brand). It's a 'Rock-a-Stack' which is neither a 'play mat' nor 'musical'.

* **Relevance Level:** **Weakly Relevant**. The connection is extremely tenuous, limited to only the brand name. It's a Fisher-Price toy, but completely unrelated to 'musical' or 'play mat'.

* **Document [9]:** 'Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.'

* **Analysis:** This document is a food recipe. It contains no keywords or conceptual links to the query.

* **Relevance Level:** **Completely Irrelevant**. No connection whatsoever.

* **Document [10]:** 'How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.'

* **Analysis:** This document explicitly mentions 'Fisher-Price musical baby play mat' and is a guide on how to clean it. It directly refers to the exact product type and brand specified in the query.

* **Relevance Level:** **Strongly Relevant**. While not a product listing, it is highly pertinent as it provides practical information about the very product the user is searching for. It confirms the existence and offers utility regarding the specific item.

Synthesizing the Ranking:

Based on the detailed analysis and comparisons, the documents naturally fall into distinct tiers of relevance:

1. **Top Tier (Completely Relevant):** Documents [1] and [6] are identical and provide direct product listings for Fisher-Price musical play mats. They offer immediate and precise fulfillment of the query.
2. **Second Tier (Strongly Relevant):** Documents [5] and [10] discuss the specific Fisher-Price musical play mat. Document [5] provides user reviews, and Document [10] offers cleaning instructions. Both are highly useful and directly pertain to the existence and characteristics of the queried product.
3. **Third Tier (Moderately Relevant):** Documents [3] and [4] partially meet the query's criteria. Document [3] is a general guide to musical play mats but lacks brand specificity. Document [4] is a Fisher-Price play mat but lacks the musical feature. They offer partial matches but miss a key component.
4. **Fourth Tier (Weakly Relevant):** Documents [7] and [8] have very limited connections. Document [7] is a Fisher-Price musical toy but not a play mat. Document [8] is merely a Fisher-Price toy, with no musical or play mat relevance.
5. **Bottom Tier (Completely Irrelevant):** Documents [2] and [9] have no relevance at all to the query's topic.

This logical progression forms the basis for the final ranking.

Here is a comprehensive assessment of the relevance of each document to the query 'fisher price musical play mat', categorized by their connection to the user's specific information need:

- * Document [1] demonstrates a complete alignment with the query's intent, presenting exact matches for products like 'Fisher-Price Kick and Play Piano Gym' and 'Fisher-Price Music and Lights Deluxe Gym'.
- * Document [2] bears no connection whatsoever to the query's subject matter.
- * Document [3] is moderately related, offering a guide on 'musical baby activity gyms' which is a direct product category, yet it omits the specific 'Fisher-Price' brand mention.
- * Document [4] is also moderately related, featuring a 'Fisher-Price Activity Mat', but crucially lacks any musical attribute.
- * Document [5] is closely related, providing customer reviews specifically for a 'Fisher-Price Deluxe Kick & Play Piano Gym', highlighting its musical features.
- * Document [6] is completely relevant, being an identical set of search results to Document [1], directly fulfilling the query with specific musical play mats by Fisher-Price.
- * Document [7] is somewhat related, as it mentions 'Fisher-Price' and 'songs and phrases', but refers to products like a 'laptop' or 'train', not a 'play mat'.
- * Document [8] is weakly related, mentioning only 'Fisher-Price' and a generic toy ('Rock-a-Stack'), with no musical or play mat relevance.

Figure B20: Example of a listwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (4/5).

- * Document [9] is entirely unrelated, discussing a recipe for lasagna.
- * Document [10] is significantly relevant, providing cleaning tips specifically for a 'FisherPrice musical baby play mat', directly addressing the queried item.

Based on this evaluation, the documents are ranked as follows:

$[6] = [1] > [10] = [5] > [3] = [4] > [8] = [7] > [9] = [2]$

Before giving the final reply to the user, let me summarize once more. Based on the above reasoning, Based on relevance scores, Document [1] has a score of 4, Document [2] has a score of 0, Document [3] has a score of 2, Document [4] has a score of 2, Document [5] has a score of 3, Document [6] has a score of 4, Document [7] has a score of 1, Document [8] has a score of 1, Document [9] has a score of 0, Document [10] has a score of 3. the final document ranking should be: $[6] = [1] > [10] = [5] > [3] = [4] > [8] = [7] > [9] = [2]$. I will output the ranking expression with scores according to the required format.
</think>

$[6](4) = [1](4) > [10](3) = [5](3) > [3](2) = [4](2) > [8](1) = [7](1) > [9](0) = [2](0)$

Figure B21: Example of a listwise fine-grained training sample in messages format with explicit reasoning (/think mode) in TFRank (5/5).

[system] Based on the relevance of the Documents to the Query and the Instruct provided to complete the task.

[user] <Instruct>: Please rank the list of documents based on their relevance to the query.

Instructions:

- Directly output the list ranking expression,
- Documents with higher relevance should be placed in front,
- Use ">" and "=" to connect document indices to indicate relevance relationships,
- In the list ranking expression, annotate each document with its relevance score based on the relevance score description, in the format [DocumentIndex](RelevanceScore).

- Relevance scores are represented by numbers from 0 to 4, with the following meanings:

0 means completely irrelevant,

1 means weakly relevant,

2 means moderately relevant,

3 means strongly relevant,

4 means completely relevant.

<Query>: fisher price musical play mat

<Documents>:

[1] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[2] - The Case Against Strict Liability. Alan Schwartz. This Article is brought to you for free and open access by FLASH: The Fordham Law Archive of Scholarship and History. It has been accepted for inclusion in Fordham Law Review by an authorized administrator of FLASH: The Fordham Law Archive of Scholarship and History.

[3] A guide to choosing the best musical baby activity gym for infants. Many brands offer mats with piano keys or sound effects.

[4] Fisher-Price Bright Beginnings Activity Mat, a soft play surface with attached arches and dangling toys for baby to reach and bat at.

[5] Customer reviews for the Fisher-Price Deluxe Kick & Play Piano Gym often highlight its engaging musical features and comfortable mat design.

[6] Search results See all results for Fisher Price Play Mat. 1 Fisher-Price Kick and Play Piano Gym by Fisher-Price. \$ 31 99. 2 Amazon's Choice for Fisher Price Play Mat. Fisher-Price Music and Lights Deluxe Gym, Rainforest by Fisher-Price. 3 Fisher-Price 3-in-1 Musical Activity Gym by Fisher-Price. More options available: \$31.99.

[7] Fisher-Price® Laugh & Learn Puppy's Smart Stages® Train Fisher-Price® Laugh & Learn® Smart Stages™ Laptop. Fisher-Price's Laugh & Learn Laptop with Smart Stages technology makes learning come to life for your baby. The interactive laptop features 9 fun keys and a space bar to press to hear songs and phrases about letters, numbers, colors, and more.

[8] Fisher-Price Brilliant Basics Rock-a-Stack. This classic toy helps babies develop hand-eye coordination and introduces them to colors.

[9] Recipe for a delicious homemade lasagna with layers of pasta, rich meat sauce, and creamy béchamel.

[10] How to clean your Fisher-Price musical baby play mat: tips for keeping it hygienic and safe for your little one.

/no think

[assistant] <think>

</think>

[6](4) = [1](4) > [10](3) = [5](3) > [3](2) = [4](2) > [8](1) = [7](1) > [9](0) = [2](0)

Figure B22: Example of a listwise fine-grained training sample in messages format without explicit reasoning (/no think mode) in TFRank.